ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · B6 · SYSTEMS AND MODELS

Inventions and where they came from

★☆☆
Grade 5

Learning goals

- Link technologies to civilisations.
- Identify parts of a system.

1 Match each technology to the civilisation it came from. [5]

Draw a line to match each pair, or write the letter.

covered drains in planned cities ●	→ E	● **A** Persia
shadoof ●	→ D	● **B** China
windcatchers and qanats ●	→ A	● **C** Bengal
magnetic compass ●	→ B	● **D** Ancient Egypt
fine muslin cloth ●	→ C	● **E** Indus Valley

2 Complete the table for a hand tube-well system. [3]

Inputs	Parts	Outputs
a push on the handle (force)	handle, rod, plunger, valves, pipe	water from underground

3 True or false? [4]

Statement	True	False	If false, correct it
1. A globe is a model of the Earth.	✔		
2. Models always show everything about the real thing.		✔	All models have limitations; they leave some things out.
3. A shadoof is a kind of lever.	✔		
4. Windcatchers need electricity to work.		✔	They use the wind.

4 Name three models you have used in school. [3]

e.g. a globe, a map, a skeleton model, a water cycle diagram, a solar system model, a paper helicopter

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · B6 · SYSTEMS AND MODELS

How did it work? Why does it matter?

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Grade 5

Learning goals

- Explain how old technologies work.
- Evaluate models.

1 **Explain how a shadoof makes lifting water easier.** [2]

It is a lever on a pivot: the heavy counterweight on one end helps lift the full bucket, so the farmer needs less force.

2 **Why were covered drains important for the health of people in Mohenjo-daro?** [2]

They carried dirty water and waste away from homes, so fewer germs spread and fewer people got diseases.

3 **Choose one model and complete the table.** [3]

Model	What it shows well	What it does not show
e.g. a globe	the shape of the Earth, continents and oceans	real size, mountains' height, weather, the Earth's movement

4 **A weather forecaster says, "Our computer model predicts the cyclone will reach the coast on Tuesday." Why do forecasts sometimes change as the storm gets closer?** [2]

The model is a prediction based on data; new measurements of the storm (from satellites) update the model, so the prediction gets more accurate.

5 **How could an old idea help solve a problem in Dhaka today? Give one example.** [2]

e.g. windcatcher towers to cool buildings without air conditioning; rainwater harvesting like old step-wells; drains like the Indus cities to stop waterlogging.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · B6 · SYSTEMS AND MODELS

Does a windcatcher model cool a room?

★★☆
Grade 5

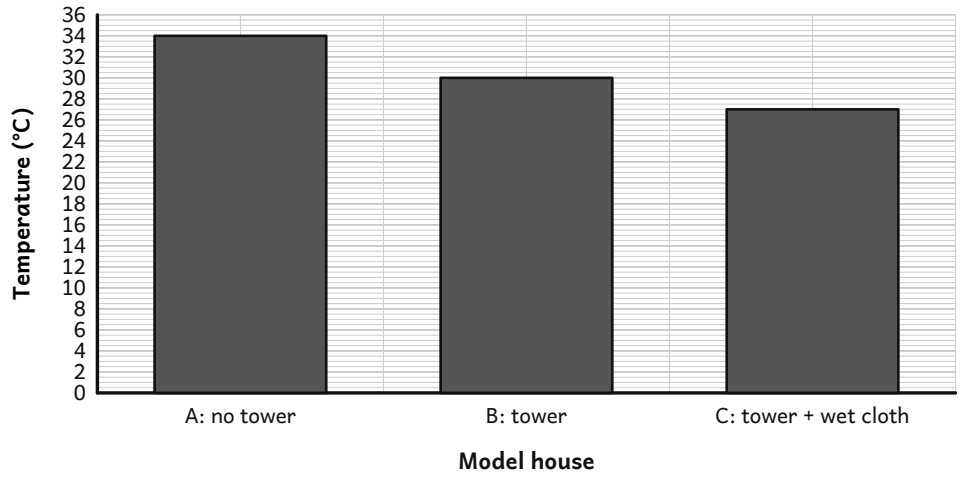
Learning goals

- Test a model fairly.
- Interpret results.
- Evaluate the model.

Class 5B made three model houses from shoe boxes. They put each one 50 cm from a desk fan and a lamp for 20 minutes and then measured the temperature inside. All boxes started at 26 °C.

Model house	Temperature after 20 minutes (°C)
A: no tower, closed	34
B: windcatcher tower facing the fan	30
C: windcatcher tower with a wet cloth inside	27

1 **Draw a bar chart of the temperatures.** [4]



2 **Which house stayed coolest? Explain why, using the words *wind* and *evaporation*.** [3]

House C: the tower brought in moving air (wind), and water evaporating from the wet cloth took heat away, cooling the air further.

3 **Name two things the class kept the same.** [2]

distance from the fan and lamp, time (20 minutes), size of box, starting temperature, thermometer

4 **Give one limitation of this model compared with a real house.** [1]

A shoe box is tiny; real wind changes direction and strength; a real house has people, sun on the roof and many rooms.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · B6 · SYSTEMS AND MODELS

Bringing back Dhaka muslin

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Grade 5

Learning goals

- Explain how a past technology connects to today.
- Consider history, science and economics together.
- Form an opinion with reasons.

Science | History | Design | Economics | Culture

For hundreds of years, weavers in the Dhaka region made **muslin**, one of the finest cloths in the world. It was woven from a special cotton plant called **phuti karpas**, which grew along the banks of the Meghna river. Spinners twisted its short fibres into threads so thin they were almost invisible, often working in the damp morning air so that the threads did not break.

Muslin was traded to kings and queens across Asia and Europe. But in the 1800s, cheap machine-made cloth from factories, changes in trade and the loss of the skilled workers caused muslin making to die out. Even the phuti karpas plant was thought to be lost.

Recently, Bangladeshi scientists and weavers searched for the plant, compared its DNA with old muslin samples in museums, grew it again and trained weavers. In 2021 they succeeded in weaving new Dhaka muslin.

Based on news reports about the Bangladesh muslin revival project.

1 **Draw a timeline of the story of Dhaka muslin with at least four events.** [3]

Timeline

centuries of muslin making → famous across Asia and Europe → 1800s decline (factory cloth, trade changes) → plant thought lost → scientists find the plant and compare DNA → 2021 new muslin woven

2 **How did scientists use evidence to find the right cotton plant?** [2]

They compared the DNA of plants they found with DNA from old muslin samples in museums to find a match.

3 **Why did spinners work in damp air? Use what you know about materials.** [2]

Damp air kept the very fine cotton fibres slightly moist and flexible, so the threads were less likely to snap.

4 **Muslin is very expensive and slow to make. Should the government keep supporting the revival? Give your opinion and two reasons.** [3]

Accept reasoned views, e.g. Yes: it protects heritage, gives skilled jobs, attracts tourism and pride. Or: money could go to other needs, so it should become self-supporting.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · B6 · SYSTEMS AND MODELS

Show what you know

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Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

D	S	F	G	I	E	W	D	G	M	B
R	Y	S	J	M	O	D	E	L	U	K
A	S	K	B	C	F	N	C	Y	S	J
I	T	I	W	O	M	P	C	S	L	C
N	E	S	V	M	H	H	S	I	I	T
F	M	J	B	P	L	P	T	O	N	M
P	H	S	N	A	E	R	V	U	G	O
C	O	G	H	S	V	P	K	T	L	G
L	U	A	T	S	E	T	R	P	O	N
B	K	T	F	S	R	H	U	U	B	K
L	F	G	J	I	N	P	U	T	E	N

compass **drain** **globe** **input** **lever** **model** **muslin** **output** **system**

2 **Which is *a* model of the Earth?** [1]

- ○ *a* compass
- ● ***a* globe**
- ○ *a* shadoof
- ○ *a* drain

3 **What does *a* windcatcher do?** [1]

- ○ lifts water
- ● **cools *a* building using the wind**
- ○ tells the time
- ○ makes electricity

4 **Design *a* model to show one system you have learned *about* this year (for example, the water cycle or the digestive system). List the materials and one limitation.** [3]

My model

Any sensible model with labelled parts, materials listed and a limitation (e.g. not to scale, does not move).

5 **Challenge. Explain why a city is a system. Name three parts and how they depend on each other.** [2]

It has many connected parts: e.g. roads, water supply, drains, electricity, people. If drains block, roads flood and transport stops.